

Periodontology & Oral Pathology

Departmental Objectives (Periodontology)

After completion of Periodontology course, undergraduate dental student will be able to:

- Define periodontology and describe the background and scope of periodontology.
- Define and describe the periodontium and their function.
- Distinguish the features of healthy and inflamed gingiva
- Classify the periodontal diseases according to severity and describe clinical features and describe their treatment plan as per needs.
- Explain the role of local and systemic factors as etiology of gingivitis and periodontitis.
- Describe the systemic disease involvements on periodontal diseases.
- Perform scaling and root planning-curettage with oral hygiene instructions with full aseptic precaution.
- Use the instruments for periodontal therapy and maintenance.
- Explain the prevention of periodontal diseases and prevention of the progress of the diseases.
- Describe basic concepts and periodontal aspects of dental implant and management of periodontal health of implanted person.
- Develop attitude for further learning of the subjects.

List of Competencies to acquire :

At the end of the course of periodontology the undergraduate students will be able to:

- Diagnose and manage common periodontal diseases.
- Execute the basic principles of periodontal instrument.
- List and identify the basic periodontal instruments.
- Perform scaling and root planning with full aseptic precautions.
- Perform gingival curettage and simple gingivectomy.
- Apply knowledge and understanding of prevention of periodontal diseases.
- Indicate required investigations for periodontal disease diagnosis.
- Perform treatment with full aseptic precaution.
- To refer the patients who require specialist's care.

Departmental Objectives (Oral Pathology)

After completion of Oral Pathology course undergraduate dental students should be able to:

- Explain basic mechanism of different types of diseases that involve the orofacial tissues.
- Describe the manifestation of common oral diseases, their diagnosis and correlate with systemic, physical signs and laboratory findings.
- Describe the oral manifestation of systemic and metabolic diseases.
- Demonstrate knowledge and understanding about the use of oral histopathology, FNAC, cytological examination, frozen section and immuno-histochemistry.
- Take a history from the patient in the specific manner
- Carry out systemic examination of the patient i.e. general examination, head and neck examination and oral cavity examination
- Diagnose common oral diseases and list differential diagnosis and construct the appropriate treatment plan for the patients.
- Describe different methods of sterilization and select appropriate methods of sterilization in the clinical practice
- Develop attitude for further learning of the subjects.

List of Competencies to acquire:

- Writing oral histopathology requisition form.
- Preservation surgical specimen and preparation of fixative for surgical specimens in 10% formalin.
- Preservation of surgical specimens for immunohistochemistry.
- Handling and maintenance of microscope.
- Interpretation of pathological reports and data.
- Writing advice for pathological investigation.
- Diagnose and manage common oral diseases and give proper counseling to patients and relatives.
- Diagnose and manage medical emergencies which may arise in dental practice.
- Demonstrate the awareness of the need to keep abreast to new knowledge and technique in oral pathology and oral medicine.

Distribution of teaching - learning hours

Lecture	Tutorial	Practical	Clinical Teaching	Total Teaching hours	Integrated Teaching (Common)	Formative Exam		Summative exam	
						Preparatory leave	Exam time	Preparatory leave	Exam time
35hrs(Oral pathology) 16hrs (Oral medicine) 34hrs (Periodontology)	24 hrs, oral path 12 hrs, oral medicine 24hrs.periodontology	26hrs,oral pathology	Dental OPD 5 Weeks(60 hrs) Periodontology, Oral medicine	85hrs, oral path 28hrs, oral medicine 118hrs, periodontology	10 hrs	10 days	20 days	10 days	25 days

Teaching – Learning Methods, Teaching Aids and Evaluation

Teaching Methods				Teaching aids	In course evaluation
Large group	Small group	Self learning	Others		
Lecture	Tutorial Practical Clinical Teaching in Dental OPD	Self study & self-assessment	Integrated Teaching	Computer & multimedia , Chalk & Board, White Board Marker, Slide Projector, Models, Surgical Specimens & Tooth Specimens, blood films	Item Examination, Card Final (Written+oral) Term final (Written +oral+ Practical+clinical)

Related Equipment:

Periodontology: Periodontal probes, Basic scaler sets, Basic root planning sets, Basic instruments for periodontal surgery, Instrument sharpening tools.

Oral Pathology

Teaching Microscope, Microscope with projection system, incubator, balance, water bath, and cell counter, computer, autoclave, haemocytometer

3rd Professional Examination:

Marks distribution of Assessment of Periodontology and Oral Pathology

Total marks – 300

- Written -100 (MCQ- 20+ SAQ- 70+ Formative Assessment 10)
- SOE-100
- Practical- OSPE and Clinical -100

Learning Objectives and Course Contents in Oral Pathology & Periodontology

Oral Pathology

Learning Objectives	Contents	Teaching Hours
Students will be able to - Describe scope and outline of oral pathology - Describe oral manifestations of systemic diseases	1. Introduction Introduction to different Pathological process involve in oral cavity. Outline the oral manifestations of systemic diseases	L – 1hr
- Describe the importance of developmental disturbances - Explain etiology, clinical features, radiology, histopathology of developmental disturbances of teeth - Describe common developmental disturbances of jaws - Explain common developmental disturbances of oral soft tissue of paraoral tissues	2. Developmental disturbances of teeth, jaws, soft tissues of oral and paraoral region - Introduction to developmental disturbances. - Developmental disturbances of teeth. - Developmental disturbances of jaws - Developmental disturbances of oral and paraoral soft tissues	L-4hrs T-4hrs
- Define dental caries - Explain microbiology of dental caries - Describe clinical features of dental caries - Describe histopathology of enamel, dentine and cementum - Describe prevention of caries	3. Dental caries Etiopathogenesis, classification, microbiology, clinical features, histopathology, prevention and its sequelae	L – 2hrs T – 2hrs P – 3hrs

Learning Objectives	Contents	Teaching Hours
Students will be able to – ● Classify pulpal diseases ● Enumerate common causes of pulpitis ● Describe clinical features of pulpitis ● Enumerate common periapical diseases and its causes ● Describe clinical features of periapical diseases ● Classify osteomyelitis of jaws ● Enumerate common causes of osteomyelitis of jaws ● Describe clinical features and histopathology and radiology of common osteomyelitis of jaws ● Explain complication of pulpitis, periapical diseases and osteomyelitis	4. Pathology of pulp, periapical tissue and Osteomyelitis Diseases of the dental pulp Diseases of the periapical tissues Osteomyelitis of the jaws	L – 3hrs T – 2hrs P – 2 hrs
● Describe common bacterial, viral and fungal infections occurring in the oral cavity ● Mention oral manifestation of bacterial, viral and fungal diseases ● Explain histopathology and laboratory diagnosis of common bacterial, viral and fungal diseases of oral cavity	5. Microbial infections of oral soft tissues Microbiology, defense mechanism including immunological aspect, oral manifestation, histopathology and laboratory diagnosis of common bacterial, viral and fungal infections.	L – 3hrs T – 1hrs P – 2hrs
● Classify non inflammatory diseases of the jaws ● Describe causes of fibro-osseous lesions of the jaws ● Describe clinical features, histopathology and radiology of common fibro-osseous lesions of the jaws ● Explain effects of common non-inflammatory diseases in oral cavity	6. Common non-inflammatory diseases of the jaws Etiopathogenesis, clinical features, radiology, histopathology and laboratory diagnosis of :Fibrous dysplasia, Cherubism, Paget's disease, osteogenesis imperfect, Rickets, Cleidocranial dysplasia, Down's syndrome, Histocytosis X disease	L – 3hrs T – 2hrs P – 2hrs

Learning Objectives	Contents	Teaching Hours
Students will be able to – ● Define ankylosis and trismus and its causes ● List types of arthritis, developmental malformations, traumatic injuries ● Explain etiology, clinical features and diagnosis of myofascial pain ● dysfunction syndrome	7. Diseases of Temporo mandibular joint Ankylosis, types of arthritis, developmental malformation, traumatic injuries and myofascial pain dysfunction syndrome	L – 1hrs T – 1hrs
● Define and classify cysts of the jaws and soft tissues ● Describe the clinical features, radiology and histopathology of common odontogenic cyst of the jaw, ● Describe clinical features, radiology and histopathology of common non-odontogenic cysts of the jaws ● Define pseudocyst ● Enumerate pseudocyst of the oral cavity ● Describe the clinical features and histopathology of common cyst occurring in oral soft tissue.	8 Cysts of the oral and paraoral region Classification, etiology, clinical features, histopathology, laboratory and radiological features odontogenic cysts, non-odontogenic cysts and pseudocysts of the jaws and soft tissues	L-4 hrs T-1hrs P-3hrs

Oral Pathology and Periodontology

Oral Pathology

Learning Objectives	Contents	Teaching Hours
Students will be able to – • Classify odontogenic tumors • Describe etiology, clinical features, histopathology, radiological features of common benign and malignant odontogenic tumours of jaws • Classify non-odontogenic tumour of the jaws • Describe etiology, clinical features, histopathology, radiological features of common benign and malignant non-odontogenic tumours of the jaws • Classify tumours of the oral epithelial tissue. • Describe clinical features and histopathology of common benign and malignant tumours of epithelial origin • Classify tumours of oral connective tissue • Describe clinical features, histopathology of common benign and malignant tumours of the oral connective tissue. • Classify salivary gland tumours. • Describe common benign and malignant tumours of the salivary glands.	9. Tumours of the oral cavity Classification of odontogenic, non-odontogenic and salivary glands tumours Etiopathogenesis, clinical features, histopathology and radiological features and laboratory diagnosis of common odontogenic, non-odontogenic and salivary gland tumours	L-10hrs T-5hrs P-7hrs
• Mention injuries to sinus • Describe dental causes and non-dental causes of sinusitis • List common cysts and tumours of maxillary sinus	10. Diseases of maxillary sinus Traumatic injuries to the sinus Sinusitis Common cysts and tumours	L – 1hrs T – 1hrs P – 2 hrs

Learning Objectives	Contents	Teaching Hours
Students will be able to – ● Define biopsy ● Explain value of biopsy ● Describe technique of biopsy ● Perform routine staining of biopsy	11. Biopsy & frozen section in diagnosis of oral diseases biopsy, value of biopsy, technique of biopsy routine staining of biopsy	L – 1hrs T – 1hrs P – 2hrs
● Mention oral manifestation of blood dyscrasia ● Describe oral aspect of disturbance in mineral metabolism ● Explain oral aspect of endocrine dysfunction	12. Systemic diseases involving oral cavity Oral manifestation, diagnosis and significance of common blood, nutritional, hormonal and metabolic diseases.	L-1 hrs T-1 hrs
● Classify common cases of oral pigmentation ● Mention causes of discoloration of teeth ● Describe clinical features and histopathology of melanin pigmentation	13. Pigmentation of Oral and paraoral region and Discoloration of Teeth Causes and Clinical manifestation	L-1 hrs T-1 hrs P-2 hrs

Periodontology and Oral pathology

Oral Medicine

Learning Objectives	Contents	Teaching Hours
Students will be able to – Explain the usefulness of oral medicine	1. Introduction usefulness of oral medicine	L-1 hr
- Classify oral ulcers - Enumerate common infective and non-infective ulcer occurring in the oral cavity - Describe clinical features, histopathology and laboratory investigation for common infective and non-infective ulcers occurring in oral cavity - Describe treatment plan and differential diagnosis of common infective and non-infective ulcers of oral cavity	2. Oral ulceration - Classification of oral ulcers - Common infective and non-infective ulcer occurring in the oral cavity - Clinical features, histopathology and laboratory investigation for common infective and non-infective ulcers occurring in oral cavity - Treatment plan and differential diagnosis of common infective and non-infective ulcers of oral cavity	L – 2hrs T –1hrs
- Classify white lesions of the oral cavity - Define premalignant lesions and condition - Define and classify leukoplakia - Describe pathogenesis, clinical features, histopathology and treatment of white lesions of the oral cavity	3. Oral white lesions - Classification of white lesions of the oral cavity - Premalignant lesions and condition - Classification of leukoplakia - Pathogenesis, clinical features, histopathology and treatment of white lesions of the oral cavity	L – 2hrs T –1hrs
- Classify oral auto-immune diseases - List oral vesiculo-bullous lesions - Describe pathogenesis, clinical features and treatment of common auto-immune diseases of oral cavity. - Describe pathogenesis, clinical features and treatment of common vesiculo-bullous lesions of the oral mucosa	4. Auto Immune diseases and vesiculo bullous lesions - Classification of oral auto-immune diseases - Oral vesiculo-bullous lesions - Pathogenesis, clinical features and treatment of common auto-immune diseases of oral cavity. - Pathogenesis, clinical features and treatment of common vesiculo-bullous lesions of the oral mucosa	L – 3hrs T –2hrs
- Describe dental and non-dental pain - Describe trigeminal neuralgia and its management - Describe facial paralysis and management	5. Orofacial pain/ Neuralgic pain dental and non-dental pain, trigeminal neuralgia and its management facial paralysis and management	L – 1hrs T –1hrs
- Describe local and systemic causes of common tongue lesions - Mention the effects of systemic diseases on tongue	6. Disease of Tongue - local and systemic causes of common tongue lesions - effects of systemic diseases on tongue	L-1hrs T-1hrs

Learning Objectives	Contents	Teaching Hours
Students will be able to – - Enumerate common traumatic and reactive lesions occurring in oral cavity - Describe common regressive lesions of the oral cavity - Explain the effects of radiation in oral cavity. - Describe healing of oral wound and complication	7. Traumatic, Reactive and Regressive lesions of oral cavity - Common traumatic and reactive lesions occurring in oral cavity - Common regressive lesions of the oral cavity - Effects of radiation in oral cavity. - Healing of oral wound and complication	L –1hrs T –2hrs
- Classify infective and non-infective diseases of the salivary glands - Describe clinical features and treatment of common viral and bacterial diseases of salivary glands - Define xerostomia - Enumerate common causes of xerostomia - Describe complication of xerostomia - Classify Sjogren's syndrome - Describe clinical features, radiology, laboratory investigation and treatment of Sjogren's syndrome	8. Non-neoplastic diseases of salivary glands - Classification of infective and non-infective diseases of the salivary glands - Clinical features and treatment of common viral and bacterial diseases of salivary glands - Xerostomia, common causes of xerostomia - Complication of xerostomia - Classification of Sjogren's syndrome, clinical features, radiology, laboratory investigation and treatment of Sjogren's syndrome	L – 2hrs T –2 hrs
- Classify red lesions of the oral cavity - Define and classify erythroplakia - Describe pathogenesis, clinical features, histopathology and treatment of red lesions of the oral cavity	9. Oral red lesions Classification of red lesions of the oral cavity potentially malignant disorders of the oral mucosa Classification of erythroplakia Pathogenesis, clinical features and treatment of oral red lesions of the oral cavity	L – 1hrs T –1 hrs
- Classify oral pigmented lesions - Describe pathogenesis, clinical features and treatment of common pigmented lesions of the oral mucosa	10. Pigmented lesions Classification of oral pigmented lesions Pathogenesis, clinical features and treatment of common pigmented lesions of the oral mucosa.	L – 1 T –1
- Define sterilization, disinfection and antisepsis - Describe certain methods of sterilization and disinfection and outline their application - Select appropriate method of sterilization in the clinical practice	11. Sterilization and Disinfection Define, classification and application of sterilization, disinfection and antisepsis Methods of sterilization: detail of autoclaving, hot air oven and chemical methods Sterilization of dental and medical equipment: Critical, Semi-critical and non-critical devices Disinfection body fluid spillage	L – 2 T –2

Oral pathology and Periodontology

Periodontology (Theory)

Learning Objectives	Contents	Teaching Hours
Students will be able to - - Define periodontology and periodontium - Describe brief history and advancement of periodontal disease status and treatment	Introduction - Definition of periodontology and periodontium - Brief history and advancement of periodontal disease status and treatment	L-1hrs
- Describe anatomy, histology and physiology of periodontium - Describe the types of gingival epithelium in relation to tooth - Describe gingival sulcus and its importance - Describe oral mucosa and explain aging of periodontium	Periodontium - Anatomy, histology and physiology of periodontium - Types of gingival epithelium in relation to tooth - Gingival sulcus and its importance - Oral mucosa and aging of periodontium	L-3hrs T-1hrs
- Describe defense mechanism of gingiva - Explain the role of saliva and gingival crevicular fluid in oral health	Oral environment for health - Defense mechanism of gingiva - Role of saliva and gingival crevicular fluid in oral health	L-1hrs T-1hrs
- Classify periodontal diseases. - Describe periodontal examination methods and Index system - Explain risk factors for gingivitis and periodontitis - Describe epidemiology of periodontal diseases	Periodontal disease - Classification of periodontal diseases. - Periodontal examination methods and Index system - Risk factors for gingivitis and periodontitis - Epidemiology of periodontal diseases	L - 4hrs T - 3hrs
- Describe and list the microorganisms involve in periodontal diseases Explain their impacts on periodontium - Describe immunological responses by periodontal tissues	Microbiology of Periodontal diseases	L-2hrs T-2hrs

Learning Objectives	Contents	Teaching Hours
Students will be able to - Describe pathogenesis and histopathological changes of periodontal tissues in specific gingival and periodontal diseases	Periodontal Pathology	L - 2hrs T - 2hrs
Explain the role of periodontic-endodontic inter-relationship in periodontal diseases.	Periodontics – Endodontics continuum	L - 1hrs T - 1hrs
- Describe chronic and acute gingival conditions (NUG). - Describe Gingival abscess - Describe desquamatic gingivitis - Enumerate the different conditions of gingival enlargement - Describe causes and types of gingival recession - Describe causes and management of tooth hypersensitivity	Gingival diseases - Chronic and acute gingival conditions (NUG). - Gingival abscess, desquamatic gingivitis - Different conditions of gingival enlargement - Causes and types of gingival recession - Causes and management of tooth hypersensitivity	L-3hrs T-2hrs
- Classify periodontitis - Define and classify periodontal pocket. - Explain mechanism of formation of pocket - Describe minor features of chronic periodontitis - Describe aggressive periodontitis - Explain refractory and necrotizing periodontitis - Describe periodontal abscess	Periodontal diseases - Classification of periodontitis - Definition and classification of periodontal pocket. - Mechanism of formation of pocket - Minor features of chronic periodontitis - Aggressive periodontitis - Refractory and necrotizing periodontitis - Periodontal abscess	L-7hrs T-4hrs

Learning Objectives	Contents	Teaching Hours
Students will be able to - List and describe cysts and tumours of the periodontium	Cysts and tumours of periodontium	L-1hrs T-1hr
Explain the effects of external forces on periodontium	Periodontal response to external forces	L-1hrs T-1hrs
Describe basic principles of periodontal instrumentation	Basic principles of periodontal instrument	L-2hrs T-2hrs
Interpret Non-surgical/Phase I /initial therapy	Phase I periodontal therapy	L-1hrs T-1hrs
- List periodontal surgical therapy - Describe different types of periodontal surgery	Surgical periodontics	L-2hrs T-1hrs
Describe periodontal regenerative technique(GTR ,GBR , bone grafts, connective tissue graft	Periodontal regenerative technique	L-1hrs T-1hrs
- Describe the periodontal aspects of dental implants - Describe periimplantitis - Explain management of periimplantitis	Dental Implantology	L-1hrs T-1hrs

Periodontology (Clinical)

Learning Objectives	Contents	Teaching Hours
Students will be able to - - Describe Symptom, Clinical assessment, Radiographic Analysis and Diagnosis of Gingivitis and Periodontitis and present the case (All data are to be recorded in prescribed form supplied by the dept)	Patient Examination, diagnosis and case presentation.	1wk
Explain the basic principles of periodontal instruments and to perform it.	Principles of periodontal instruments	1wk
Furnish initial treatment plan and treat gingivitis and periodontitis cases.	Treatment Planning	1wk
- Provide oral hygiene instruction to the patients - Perform Scaling and polishing - Perform root planning	Phase 1 Periodontal Therapy	1wk
Describe General guidelines for periodontal Surgery; - Resective periodontal surgery to eliminate gingival overgrowth and/or pocket—Gingivectomy and gingivoplasty - Periodontal flap surgery: indication and rationality, Techniques of different procedures; ENAP,MWFP,IBIFP etc - Occlusal adjustment, osseous recontouring splinting for stabilization of vulnerable mobile but sound tooth, - Mucogingival surgery: management of Gingival recession, gingival augmentation procedure, Root coverage procedure(sliding flap and envelop flap), Treatment of furcation-involved tooth(tunneling and/or reposition flap)	Phase II/ Corrective Periodontal Therapy General guidelines for periodontal Surgery; - Resective periodontal surgery to eliminate gingival overgrowth and/or pocket—Gingivectomy and gingivoplasty - Periodontal flap surgery: indication and rationality, Techniques of different procedures; ENAP,MWFP,IBIFP etc - Occlusal adjustment, osseous recontouring splinting for stabilization of vulnerable mobile but sound tooth, - Mucogingival surgery: management of Gingival recession, gingival augmentation procedure, Root coverage procedure(sliding flap and envelop flap), Treatment of furcation-involved tooth(tunneling and/or reposition flap)	1wk
Enumerate local and systemic antibacterial drugs used in periodontal therapy.	Chemo-therapeutic agents in periodontal therapy, Pocket Irrigation Systemic administration	1hr
Interpret Supportive Periodontal therapy	Phase- 3/ Maintenance and recall therapy /SPT Re-examination, Prophylaxis, Monitoring of OHI	1hr
Decide cases for referral to specialist	Interdisciplinary referral system	1hr

Summative Assessment of Oral Pathology & Periodontology

(3rd Professional Examination)

Assessment System and Mark Distribution

Components	Marks	Total Marks
Formative Assessment	10	10
Written Examination Group-A: Oral Pathology & Oral Medicine : (MCQ+SAQ) Group-B: Periodontology: (MCQ+SAQ)	(10+35) (10+35)	90
Practical +Clinical Examination OSPE Clinical Practical Note Book	40 50 10	100
Oral Examination (Structured) 1 Board Oral Pathology Oral Medicine Periodontology	40 20 40	100

- Pass marks 60% in each of theoretical, oral, practical and clinical
- Oral, Clinical & Practical Examination will be in 2 days
 - One day-OSPE
 - another day-Oral & Clinical